

MATHEMATICS
Introduction to Modern Algebra II
Quiz 3.

1) Show that $x^3 + 3x + 2$ is irreducible in $\mathbb{Q}[x]$. Is it also irreducible in $\mathbb{R}[x]$?

Proof. We know that $f(x) = x^3 + 3x + 2$ is irreducible if and only if so is $f(x+1) = (x+1)^3 + 3(x+1) + 2 = x^3 + 3x^2 + 6x + 6$. By Eisenstein's Criterion (with $p = 3$), $f(x+1)$ is irreducible.

$x^3 + 3x + 2$ is not irreducible in $\mathbb{R}[x]$. In fact as $f(x)$ is continuous and $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$, it is easy to see that $f(x)$ has a real root and hence a linear factor in $\mathbb{R}[x]$. $\square$

2) Let F be a field and $f(x) \in F[x]$. When is $f(x)$ irreducible (i.e. give the definition). Show that if $f(x)$ is irreducible, then $(f(x))$ is a maximal ideal.

Proof. $f(x)$ irreducible if $\deg f(x) > 0$ and given any $g(x) \in F[x]$, either $f(x)$ divides $g(x)$ or $(f(x), g(x)) = 1$. (Equivalently $f(x)$ irreducible if $\deg f(x) > 0$ and if $f(x) = a(x)b(x)$ for some $a(x), b(x) \in F[x]$, then $\deg a(x) = 0$ or $\deg b(x) = 0$.)

Suppose that $f(x)$ is irreducible and $N \supset (f(x))$ is an ideal such that $N \neq (f(x))$. Let $g(x) \in N - (f(x))$, then $f(x)$ does not divide $g(x)$ and so $(f(x), g(x)) = 1$ and thus we may write $1 = a(x)f(x) + b(x)g(x)$ for some $a(x), b(x) \in F[x]$. Since $f(x), g(x) \in N$, then $a(x)f(x), b(x)g(x) \in N$ and so $1 \in N$ i.e. $N = F[x]$. $\square$